Remark 1. It was given as an exercise in class to define the semisimple and nilpotent parts of an element $x \in \mathfrak{g}$, we do that here.

Proof. The proof uses two results:

Proposition 1 (Humphreys Lemma 4.2B)**.** Let $\mathfrak{U}$ be a finite dimensional k-Algebra. Then for $x \in \mathfrak{U}$ the jordan decompoisition $x = x_s + x_n$ in $\mathrm{End}(\mathfrak{U}) \supset \mathrm{Der}(\mathfrak{U})$ satisfies $x_n, x_s \in \mathrm{Der}(\mathfrak{U})$.

Proposition 2 (Humphreys Theorem 5.3)**.** If $\mathfrak{g}$ is semisimple then $\mathrm{ad}(\mathfrak{g}) = \mathrm{Der}(\mathfrak{g})$.

We are now ready to define the semisimple and nilpotent part of an element x. Recall that $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$, then for $x \in \mathfrak{g}$ it follows that $\mathrm{ad}(x)$ has a Jordan decomposition. Now since $\mathfrak{g}$ is semisimple, we know that $\mathfrak{z}(\mathfrak{g}) \subset \mathrm{Rad}(\mathfrak{g}) = 0$ and thus $\mathrm{ad} : \mathfrak{g} \xrightarrow{\cong} \mathrm{ad}(\mathfrak{gl}(\mathfrak{g}))$. Thus we define the Jordan decomposition as follows

$$x_s = \mathrm{ad}^{-1}(\mathrm{ad}(x)_s) \text{ and } x_n = \mathrm{ad}^{-1}(\mathrm{ad}(x)_n) \tag{1}$$

The fact that $\mathrm{ad}(x)_s, \mathrm{ad}(x)_n \in \mathrm{ad}(\mathfrak{g})$ follows from Proposition 1 and Proposition 2. □

Exercise 1 (Humphrey's 5.8)**.** Let $\mathfrak{g}$ be a Lie algebra and $\bigoplus I_j$ be its decomposition into simple ideals. Let $\pi_j : \mathfrak{g} \to I_j$ be the projection, then show that

$$x_n = \sum_j \pi_j(x)_n \text{ and } x_s = \sum_j \pi_j(x)_s \tag{2}$$

Before beginning the proof it is worth noting that this decomposition uses the forward implication of $\mathfrak{g}$ is semisimple $\iff$ κ is nondegerate. This requires that the base field k have characteristic zero since the proof uses Cartan's criterion which relies on characteristic zero. A number of the results stated in defining the semisimple nilpotent decomposition also require certain niceties of the base field, for simplicity I assume that k is an algebraically closed field of characteristic zero, all of the stated propositions hold in this context.

Proof. Since $\mathfrak{g}$ is semisimple we find that for $i \neq j$ we have $[I_j, I_i] \subset I_j \cap I_i = 0$ so that

$$[\mathrm{ad}(\pi_j(x)), \mathrm{ad}(\pi_i(x))] = \mathrm{ad}([\pi_j(x), \pi_i(x)]) = \mathrm{ad}(0) = 0 \tag{3}$$

So all of the $\mathrm{ad}(\pi_j(x))$ commute. We use a fact from linear algebra that the sum of commuting semisimple (resp. nilpotent) elements is semisimple (resp. nilpotent). It follows that

$$\mathrm{ad}(x) = \sum_j \mathrm{ad}(\pi_j(x)) = \sum_j \mathrm{ad}(\pi_j(x))_s + \mathrm{ad}(\pi_j(x))_n = \underbrace{\sum_j \mathrm{ad}(\pi_j(x))_s}_{\text{semisimple}} + \underbrace{\sum_j \mathrm{ad}(\pi_j(x))_n}_{\text{nilpotent}} \tag{4}$$

Then by uniqueness of the Jordan decomposition, we find that

$$\mathrm{ad}(x)_s = \sum_j \mathrm{ad}(\pi_j(x))_s \text{ and } \mathrm{ad}(x)_n = \sum_j \mathrm{ad}(\pi_j(x))_n \tag{5}$$

It follows that

$$x_s = \mathrm{ad}^{-1}(\mathrm{ad}(x)_s) = \mathrm{ad}^{-1}\left(\sum_j \mathrm{ad}(\pi_j(x))_s\right) = \sum_j \mathrm{ad}^{-1}(\mathrm{ad}(\pi_j(x))_s) = \sum_j \pi_j(x)_s \tag{6}$$

$$x_n = \mathrm{ad}^{-1}(\mathrm{ad}(x)_n) = \mathrm{ad}^{-1}\left(\sum_j \mathrm{ad}(\pi_j(x))_n\right) = \sum_j \mathrm{ad}^{-1}(\mathrm{ad}(\pi_j(x))_n) = \sum_j \pi_j(x)_n \tag{7}$$

□